

PRAIRIE DU CHIEN, WI, USA

WMO: 726444

Lat: 43.019N

Lon: 91.124W

Elev: 202

StdP: 98.93

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04963

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-19.1	-26.9	0.3	-21.7	-24.2	0.4	-18.7	11.0	-7.7	9.5	-5.9	1.5	280	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	32.4	23.8	30.8	22.3	29.0	21.7	25.1	30.2	24.0	28.7	23.1	27.5	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.9	28.1	22.5	17.7	26.9	21.6	16.7	26.0	77.6	29.9	73.1	28.8	69.4	27.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.0	7.1	-26.5	34.6	3.1	1.5	-28.7	35.6	-30.5	36.5	-32.2	37.3	-34.5	38.3
			-26.7	26.9	3.1	1.5	-28.9	28.0	-30.7	28.9	-32.5	29.7	-34.7	30.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-6.1	-5.0	2.3	9.3	15.3	20.6	22.8	21.5	17.9	10.6	3.8	-3.2
	DBStd	11.46	7.08	6.62	6.64	5.37	4.77	3.38	3.15	3.04	4.29	5.11	5.60	6.18
	HDD10.0	1921	500	419	253	76	9	0	0	0	2	54	198	409
	HDD18.3	3736	758	653	499	275	118	15	3	6	59	245	436	667
	CDD10.0	1636	0	0	13	54	174	318	397	356	237	73	12	1
	CDD18.3	409	0	0	1	4	25	83	142	103	45	6	0	0
	CDH23.3	3621	0	0	9	49	249	746	1278	848	389	52	1	0
	CDH26.7	1076	0	0	1	9	66	219	433	238	102	9	0	0
Wind	WSAvg	2.6	2.8	2.8	3.0	3.4	2.9	2.3	1.9	1.7	2.1	2.6	2.8	2.7
Precipitation	PrecAvg	811	26	28	49	84	102	105	90	104	82	56	54	32
	PrecMax	1213	75	83	132	171	296	254	186	259	199	143	150	75
	PrecMin	518	2	1	0	26	20	14	14	9	3	0	3	5
	PrecStd	157	16	19	30	38	49	63	46	61	54	31	35	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	13.7	24.8	27.9	32.0	33.3	35.0	33.8	32.4	27.9	21.5	14.1
		MCWB	5.9	8.9	16.7	17.4	22.3	23.6	25.5	24.3	22.7	19.6	14.2	11.8
	2%	DB	7.4	9.1	18.6	24.5	28.1	31.3	32.4	31.2	29.3	24.5	17.6	9.3
		MCWB	4.3	6.0	13.4	15.6	18.6	22.2	24.6	23.0	21.5	17.3	12.5	7.0
	5%	DB	4.3	6.1	15.3	21.4	26.2	29.0	30.7	29.0	27.5	22.1	15.1	7.0
		MCWB	2.1	3.4	10.8	13.4	17.7	20.9	23.0	22.0	20.0	16.5	10.5	4.9
	10%	DB	2.7	3.7	12.0	18.7	23.9	27.5	28.9	27.7	25.5	19.1	12.5	4.7
		MCWB	1.1	1.4	7.9	11.6	16.7	20.2	22.3	21.3	18.8	14.4	8.5	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	9.2	17.6	19.1	23.1	25.4	27.5	26.0	24.3	21.7	15.5	12.8
		MCDB	10.2	13.6	24.4	25.5	28.6	30.7	32.5	31.4	29.4	26.1	19.5	14.7
	2%	WB	4.4	6.2	14.8	16.4	20.8	23.9	25.6	24.5	22.7	19.4	13.5	7.5
		MCDB	7.1	8.6	17.9	22.5	26.0	29.2	30.2	29.0	27.4	22.8	16.9	8.4
	5%	WB	2.5	3.5	10.8	14.7	19.3	22.7	24.3	23.5	21.5	17.1	11.4	4.8
		MCDB	4.2	5.8	14.6	20.0	24.0	27.8	29.2	27.3	25.6	20.8	14.4	6.8
	10%	WB	1.2	1.7	7.9	12.6	17.9	21.4	23.3	22.5	20.3	14.9	8.9	2.9
		MCDB	2.7	3.5	11.6	17.9	22.5	25.8	27.9	26.3	24.0	18.8	12.3	4.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	9.5	10.9	11.2	10.6	10.4	10.5	11.4	10.5	8.9	7.8
		MCDBR	9.7	11.2	13.4	15.3	13.6	12.6	11.7	11.9	12.9	13.6	12.6	10.0
		MCWBR	7.4	8.5	8.4	8.2	6.3	5.9	5.3	5.5	6.1	7.6	8.5	8.1
	5% WB	MCDBR	9.0	10.9	12.6	12.8	11.4	10.8	10.4	9.9	10.6	11.5	11.6	9.5
		MCWBR	7.1	8.5	8.3	8.0	6.4	6.2	5.5	5.5	5.8	8.0	8.8	8.2
Clear-Sky Solar Irradiance	taub	0.277	0.301	0.331	0.375	0.403	0.425	0.434	0.417	0.385	0.338	0.297	0.278	
	taud	2.381	2.320	2.377	2.297	2.272	2.249	2.226	2.275	2.342	2.471	2.507	2.430	
	Ebn at Noon	865	896	911	891	871	849	837	839	841	847	840	832	
	Edh at Noon	78	98	106	125	132	135	137	126	110	85	69	68	
All-Sky Solar Radiation	RadAvg	1.79	2.82	3.81	4.58	5.43	6.03	6.13	5.30	4.36	2.81	1.86	1.37	
	RadStd	0.19	0.28	0.42	0.40	0.40	0.48	0.35	0.36	0.34	0.36	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-3.96	N/A	N/A	-0.70	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page